CardioNex AI — Specific AFib & AFL Clinical Screening Report

Dedicated Abnormality Detection: Atrial Fibrillation (AFib) & Atrial Flutter (AFL)

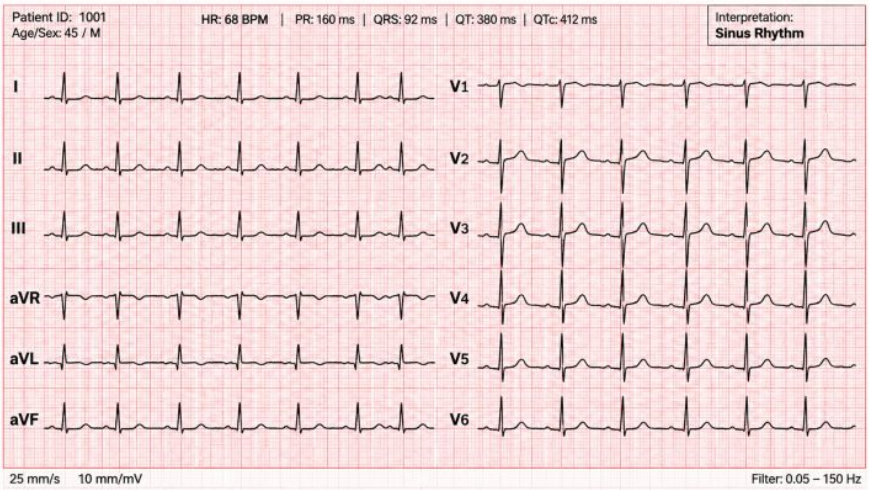
Clinical Screening Disclaimer: This tool is an AI-assisted ECG screening and decision-support system. It does not provide a definitive medical diagnosis or prescribe medications. All flagged rhythms and uncertain findings require review by a licensed physician or cardiologist.

Detection Result:	**Neither AFib nor AFL (Screening Only)**
Report Result Type:	AFib/AFL screening result
Model Architecture:	ECG Intelligence 1D Transformer (v1.0.0-clinical-cpu)
Model Probability:	**97.0%** (Temperature-calibrated T=2.061; separate from clinical certainty)
Signal Quality Assessment:	**Good** (77% signal fidelity; Continuous digital trace)

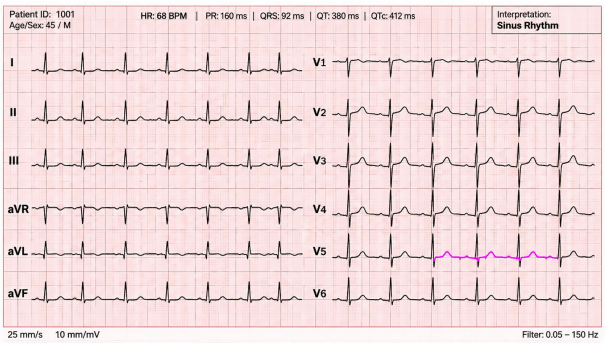
1. Report & Patient Submitted Context

Report ID	ba5811b73b15420b92e3a1b4a8c63a38
Patient ID	Not provided
Filename	Clinical 12-Lead ECG Printout-4.png
Report Generated (UTC)	2026-11-05T06:31:47.320711Z
Hospital / Clinic	Not provided
Test Location	Not provided
Patient Age	Not provided
Reported Symptoms	Not provided
Medical History	Not provided
Current Medications	Not provided
Referring Physician	Not provided

2. Original Uploaded ECG Document



3. Extracted ECG waveform preview



4. Detailed AFib / AFL Rhythm Analysis & Class Probabilities

Calculated Ventricular Rate	45.9 bpm (60 / mean RR)
Source Document Rate	68 bpm (source report)
Rate Correlation & Delta	Discrepancy detected (Waveform: 45.9 bpm vs. Source: 68.0 bpm), characteristic of marked ventricular irregularity.
QRS Complexes Detected	7 detected complexes (6 RR intervals)
RR Interval Metrics	Mean: 1308.3 ms | Median: 1040.0 ms | Range: 290.0–2520.0 ms
RR Irregularity (CV_RR)	Irregularly Irregular — CV: 0.649 (Threshold >0.15 denotes marked irregularity)
Rate Calculation Method	Pan-Tompkins QRS & R-Peak Detection (5-15 Hz Bandpass, Derivative, Squaring, MWI) (Reliability: High)
Class Probabilities	AFib: 2.1% | AFL: 0.9% | Neither: 97.0%
Dominant Flutter Frequency	4.0 Hz (Supplementary spectral check only; does not confirm AFL)
Specific Observed Evidence	• Marked RR interval irregularity (CV: 0.65, > 0.15 threshold) • Absence of consistent, discrete P waves before QRS complexes

Source Document Findings & Clinical Comparison

Heart rate reported in source document: 68 bpm (source report)

Source Impression: Age/Sex: 45 / M Sinus Rhythm | | | | | | jer | ah } al | 1a 3 E be | i ja | I | | aVF | i | E Ae | = | i | it Be eit. STR (leg (Rae (a a 25mm/s 10mm/mV Filter: 0.05 — 150 Hz

5. Technical Model & Processing Specifications

Model Architecture & Version	ECG Intelligence 1D Transformer (v1.0.0-clinical-cpu)
Training Dataset & Partition	PTB-XL Diagnostic ECG Benchmark v1.0.3
Signal Extraction & Sampling	1,000 samples @ 100 Hz (default) across 0.68 seconds
Preprocessing & Filtering	Otsu binarization, morphological filtering, 0.5–40 Hz bandpass
Attribution Explainability Method	Integrated Gradients (50 approximation steps)
Doctor Review Status	NOT_REQUESTED

6. Model & Dataset Limitations Disclosure

AFL Dataset Scarcity Disclosure	PTB-XL contains 56 AFL cases vs 1,500+ AFib cases. Model AFL test F1 is 0.462 (AFib F1: 0.942). Dominant ~4 Hz spectral energy is supplementary evidence only.
Single-Lead Extraction Limitation	Single-lead rhythm screening (Lead II) cannot evaluate complete cardiac electrophysiology. Full 12-lead ECG review is required.
Clinical Decision Safeguard	This tool is an AI-assisted ECG screening and decision-support system. It does not provide a definitive medical diagnosis or prescribe medications. All flagged rhythms and uncertain findings require review by a licensed physician or cardiologist.

7. Biomedical Explanation (RAG Grounded in Guidelines)

Clinical Context: The screening was inconclusive due to insufficient signal quality, uncalibrated image scale, or ambiguous waveform features.

Pathophysiology: The system cannot reliably confirm or exclude abnormal rhythm because the recording quality or grid calibration does not meet the strict technical criteria.

Recommended Clinical Evaluation: Repeat the ECG using a high-quality standard 12-lead digital device or obtain a clean, flat, high-contrast scan with visible grid lines.

EMERGENCY WARNING: If the patient experiences acute chest pain, shortness of breath, dizziness, syncope, or hemodynamic instability, seek immediate emergency medical care.

Biomedical Citations:

• **MedlinePlus, U.S. National Library of Medicine:** Atrial fibrillation (https://medlineplus.gov/atrialfibrillation.html)

• **MedlinePlus, U.S. National Library of Medicine:** Atrial flutter (https://medlineplus.gov/atrialflutter.html)